

Osmosis and Onsager coupling across a membrane

Deriving the Kedem–Katchalsky form

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A membrane can couple water flow to solute transport. Writing two independent permeability laws misses that coupling and can hide thermodynamic inconsistencies. Here the coupled laws are constructed from conjugate forces and fluxes, then rewritten in experimentally useful variables. The system is passive, isothermal, planar, and close to equilibrium.

1 Fix the direction before writing a law

A stationary membrane separates well-mixed left and right reservoirs of an ideal dilute solution containing one neutral solute. Set $\Delta p = p_L - p_R$ and $\Delta c = c_L - c_R$. The solute concentrations c_L, c_R are in mol/m^3 . Positive volume flux J_v and molar solute flux J_s point from left to right. Let $\bar{c} = (c_L + c_R)/2 > 0$ be the reference concentration. The temperature is T , the gas constant is R , and the osmotic pressure difference is $\Delta\pi = RT\Delta c$ at this order.

For a dilute incompressible solvent with partial molar volume $\bar{v}_w$, the solvent chemical potential satisfies

$$\Delta\mu_w \simeq \bar{v}_w(\Delta p - \Delta\pi). \quad (1)$$

Neglecting the solute contribution to the volume flux, $J_v \simeq \bar{v}_w J_w$, where J_w is the solvent molar flux. Also neglecting the dilute solute's pressure contribution,

$$\Delta\mu_s = RT \log(c_L/c_R) \simeq \frac{RT\Delta c}{\bar{c}} = \frac{\Delta\pi}{\bar{c}}. \quad (2)$$

These approximations require small relative concentration differences and a dilute solution. For concentrated solutions, activities and partial molar volumes must be retained.

The dissipated free energy per area per time, or temperature times the interfacial entropy-production rate $\dot{s}_i$, is

$$T\dot{s}_i = J_w\Delta\mu_w + J_s\Delta\mu_s \simeq J_v(\Delta p - \Delta\pi) + J_s\Delta\mu_s. \quad (3)$$

Each product has units W/m^2 . Equation (3) identifies the conjugate force–flux pairs: J_v with $\Delta p - \Delta\pi$, and J_s with $\Delta\mu_s$. A coefficient matrix can only be tested for Onsager symmetry after this pairing has been fixed.

2 Linear response and its constraints

Near equilibrium, write

$$\begin{pmatrix} J_v \\ J_s \end{pmatrix} = \begin{pmatrix} L_{vv} & L_{vs} \\ L_{sv} & L_{ss} \end{pmatrix} \begin{pmatrix} \Delta p - \Delta\pi \\ \Delta\mu_s \end{pmatrix}. \quad (4)$$

Under microscopic reversibility, with no magnetic field or other time-reversal-breaking drive in this passive transport problem, reciprocity gives $L_{vs} = L_{sv}$ [1]. The coefficients have different units on different rows; nevertheless L_{vs} and L_{sv} have the same units because their forces are conjugate.

The second law requires the symmetric transport matrix to be positive semidefinite:

$$L_{vv} \geq 0, \quad L_{ss} \geq 0, \quad L_{vv}L_{ss} - L_{vs}^2 \geq 0. \quad (5)$$

These inequalities and reciprocity are distinct requirements. Nonnegative entropy production alone does not prove equality of cross coefficients. We assume $L_{vv} > 0$ below so that water permeability is nonzero.

Define the hydraulic permeability L_p , reflection coefficient σ , and solute permeability P_s by

$$L_p = L_{vv}, \quad \sigma = 1 - \frac{L_{vs}}{\bar{c}L_{vv}}, \quad P_s = \frac{RT}{\bar{c}} \left(L_{ss} - \frac{L_{vs}^2}{L_{vv}} \right). \quad (6)$$

Their units are m/(Pa s), dimensionless, and m/s, respectively. Substitution into the first row of Eq. (4) gives

$$\boxed{J_v = L_p(\Delta p - \sigma\Delta\pi)}. \quad (7)$$

Next eliminate $\Delta p - \Delta\pi$ from the second row. Since

$$\Delta p - \Delta\pi = \frac{J_v}{L_p} - (1 - \sigma)\Delta\pi, \quad (8)$$

the solute flux becomes

$$\boxed{J_s = P_s\Delta c + (1 - \sigma)\bar{c}J_v}. \quad (9)$$

Equations (7) and (9) are the dilute linear Kedem–Katchalsky form [2]. Their two solute contributions are diffusion and solvent drag. The factor multiplying solvent drag is tied to the osmotic term in the water flux by reciprocity; it is not an independent fit parameter in this model.

3 A direct second-law check

Rewrite Eq. (3) using $\Delta\mu_s = \Delta\pi/\bar{c}$:

$$T\dot{s}_i = J_v(\Delta p - \sigma\Delta\pi) + [J_s - (1 - \sigma)\bar{c}J_v] \frac{\Delta\pi}{\bar{c}} \quad (10)$$

$$= \frac{J_v^2}{L_p} + \frac{P_s RT}{\bar{c}} (\Delta c)^2 \geq 0. \quad (11)$$

The cross terms cancel exactly. This calculation checks both the sign convention and the transformed coefficients. It also shows that $P_s \geq 0$ follows from the determinant condition in Eq. (5).

The commonly used interval $0 \leq \sigma \leq 1$ applies to familiar simple membrane models, but is not a consequence of Eq. (11) alone. The second law constrains dissipation; stronger bounds require assumptions about the transport mechanism. Likewise $\sigma = 1$ removes solvent drag in Eq. (9), but does not force $P_s = 0$. A perfectly semipermeable membrane impermeable to this solute has both $\sigma = 1$ and $P_s = 0$.

4 Stall, equilibrium, and an example

At equal hydrostatic pressure and $\Delta c > 0$, Eq. (7) predicts $J_v < 0$ for positive σ : water enters the more concentrated left side. Solute diffusion points the other way. The net solute flux depends on the competing drag contribution, so its direction cannot be inferred from Δc alone.

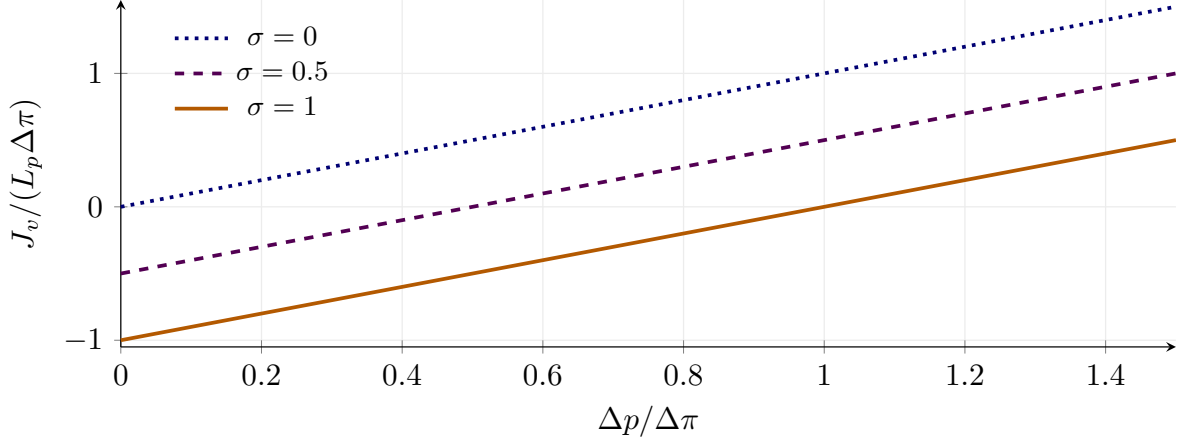


Figure 1: Water flux at fixed positive osmotic pressure difference. Each line follows Eq. (7); changing the reflection coefficient shifts the zero-flow pressure. These are illustrative constitutive curves, not measurements.

For example, let $T = 298\text{ K}$, $c_L = 110\text{ mol/m}^3$, $c_R = 100\text{ mol/m}^3$, $\Delta p = 0$, $L_p = 10^{-12}\text{ m/(Pa s)}$, $P_s = 10^{-7}\text{ m/s}$, and $\sigma = 0.8$. Then

$$\begin{aligned} \Delta \pi &= 24.8\text{ kPa}, & J_v &= -1.98 \times 10^{-8}\text{ m/s}, \\ P_s \Delta c &= 1.00 \times 10^{-6}\text{ mol/(m}^2\text{s)}, & (1 - \sigma) \bar{c} J_v &= -4.16 \times 10^{-7}\text{ mol/(m}^2\text{s)}. \end{aligned}$$

Thus $J_s = 5.84 \times 10^{-7}\text{ mol/(m}^2\text{s)}$: solute moves left to right while water moves right to left. Equation (11) gives positive dissipation, approximately $6.29 \times 10^{-4}\text{ W/m}^2$. These are illustrative linear-response parameters, not measured permeabilities of a particular cell.

The water flux stalls at $\Delta p = \sigma \Delta \pi$. This is generally not equilibrium: Eq. (9) still gives $J_s = P_s \Delta c$. For a strictly positive-definite transport matrix with both pathways available, vanishing entropy production requires both conjugate forces to vanish. If solute cannot cross, a nonzero concentration difference may instead persist at constrained osmotic equilibrium.

5 Finite reservoirs and deformable membranes

For membrane area A , reservoir volumes V_L, V_R , and solute amounts N_L, N_R , conservation requires

$$\dot{V}_L = -AJ_v, \quad \dot{V}_R = AJ_v, \quad \dot{N}_L = -AJ_s, \quad \dot{N}_R = AJ_s, \quad c_L = N_L/V_L, \quad c_R = N_R/V_R. \quad (12)$$

This closes neither the pressure nor the mechanics. One must also specify how pressure changes with volume, and whether the membrane area changes. Even with constant permeabilities the forces evolve as water and solute move.

For a curved, moving membrane, stress, geometry, surface transport, and permeation must be treated together. Sahu, Sauer, and Mandadapu develop a curved-membrane irreversible-thermodynamic framework [3]; Alkadri and Mandadapu extend the treatment to permeability and osmosis [4]. The planar derivation above is a useful check on signs and dissipation, but omits curvature and viscous-stress contributions to the interfacial forces. ATP-driven pumps also require chemical work in the entropy balance; adding a passive permeability does not account for that energy supply.

References

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